PERFECTNESS OF A MORPHISM IS FPPF LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 17997-18013, immediately following the statement at 17991-17995.

Editorial note. Three cross-chapter pointers (Descent, More on Algebra) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The statement is a single-property locality claim; the two halves of perfectness are handled separately inside the span.

Allowed standard prerequisites (editorial).

- Pseudo-coherence is fppf local on the source. chapter lemma lemma-pseudo-coherent-fppf-local-source, cited in the proof. Supplies the pseudo-coherence half of the fppf locality argument.
- Flatness of a flat locally finitely presented base change. Descent, lemma descent-lemma-flat-finitely-presented-permanence, cited in the proof. Keeps flatness along the fppf coverings that appear in the reduction.
- Tor dimension modulo a nilpotent and its base change behaviour. More on Algebra lemmas on tor amplitude cited in the proof. Converts the stalkwise tor amplitude condition under the covering into the condition on X.

Source reference. Original lemma 069C: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17901–17901. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0699: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17621–17621. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 068S: focused-seeds/geometry/S9-morph-perfect-fppf-local-source/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is perfect" is fppf local on the source.*

Proof. Let $\{g_i : X_i \to X\}_{i \in I}$ be an fppf covering of schemes and let $f : X \to S$ be a morphism such that each $f \circ g_i$ is perfect. By Lemma 0699 we conclude that f is pseudo-coherent. Hence by Lemma 069C it suffices to check that $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{S,f(x)}$-module of finite tor dimension for all $x \in X$. Pick $i \in I$ and $x_i \in X_i$ mapping to x. Then we see that $\mathcal{O}_{X_i,x_i}$ has finite tor dimension over $\mathcal{O}_{S,f(x)}$ and

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that $\mathcal{O}_{X,x} \to \mathcal{O}_{X_i,x_i}$ is faithfully flat. The desired conclusion follows from More on Algebra, Lemma 068S. $\square$